

Ch 3: Metals and Non-metals

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION A - INTRODUCTION, DEFINITIONS, OCCURRENCE AND BASIC COMPARISON

1. Introduction

- Metals and non-metals are the two broad classes of elements.
- They have different physical properties and chemical behaviours.
- Metals are widely used in construction, transport, machinery, electrical appliances and many industries.
- Non-metals are essential for life (e.g., oxygen for breathing) and used in medicines, fertilisers, fuels and many products.

2. Definition of metals

- Metals are elements that are lustrous, good conductors of heat and electricity, malleable, ductile and sonorous, and generally tend to lose electrons to form positive ions.

3. Definition of non-metals

- Non-metals are elements that are dull (non-lustrous), poor conductors of heat and electricity, usually brittle (if solid), not malleable or ductile, non-sonorous, and generally tend to gain electrons to form negative ions.

4. Metalloids (Semi-metals)

- Some elements show properties intermediate between metals and non-metals. These are called metalloids.
- They can behave like metals under some conditions and like non-metals under other conditions.
- Examples:** Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te), Polonium (Po).

5. Common examples in daily life

- Metals:** Iron, Copper, Aluminium, Gold, Silver, Zinc, Sodium.
- Non-metals:** Oxygen, Nitrogen, Carbon, Sulphur, Chlorine, Phosphorus, Iodine.

6. Occurrence in nature

- Some metals occur in free (native) state in nature because they are less reactive (e.g., Gold, Silver, Platinum, Copper).
- Many metals occur in the form of compounds (ores) and must be extracted.
- Most non-metals occur in nature as gases (e.g., O_2 , N_2 , Cl_2) or in the form of compounds (e.g., H_2O , CO_2 , SO_2 , $NaCl$).

Property	Metals	Non-metals
Appearance	Lustrous (shiny)	Dull (non-lustrous)
Physical state (at room temperature)	Mostly solids (except mercury – liquid)	Many are gases; some solids; one liquid (bromine)
Conductivity	Good conductors of heat and electricity	Poor conductors of heat and electricity (except graphite)
Malleability	Malleable – can be beaten into thin sheets	Not malleable – brittle if solid
Ductility	Ductile – can be drawn into wires	Not ductile – cannot be drawn into wires
Sonority	Sonorous – produce a ringing sound when struck	Non-sonorous – do not produce ringing sound
Reaction tendency	Generally reactive; combine with oxygen to form basic oxides	Generally less reactive; combine with oxygen to form acidic oxides
Electron behaviour	Tend to lose electrons and form positive ions (cations)	Tend to gain electrons and form negative ions (anions)

7. Examples from daily life

Iron nail (Metal)	Copper wire (Metal)	Aluminium foil (Metal)	Coal (Non-metal)	Sulphur (Non-metal)	Iodine (Non-metal)	Oxygen (Non-metal)
						
Used in construction and fixing objects.	Used in electrical wiring due to high conductivity.	Used for packing food and in kitchen applications.	Used as a fuel in industries and power plants.	Used in making sulphuric acid and in match sticks.	Used as an antiseptic and in iodised salt.	Essential for breathing and for burning.

8. Key Idea



- Metals tend to lose electrons during reactions.
→ They form positive ions (cations).
Example: $Na \rightarrow Na^+ + e^-$
- Non-metals tend to gain electrons during reactions.
→ They form negative ions (anions).
Example: $Cl + e^- \rightarrow Cl^-$



9. Exam Tips - Centum Focus

- Learn the key differences between metals and non-metals (table above).
- Remember exceptions: Mercury is a liquid metal; Bromine is a liquid non-metal; Graphite (carbon) conducts electricity.
- Know the occurrence: native metals vs metals in compounds; non-metals usually as gases/compounds.
- Relate properties to uses in daily life – helps in scoring application questions.
- Practice balanced equations for reactions of metals and non-metals.

Ch 3: Metals and Non-metals

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION B - PHYSICAL PROPERTIES OF METALS AND NON-METALS

- Physical properties can be observed or measured without changing the chemical identity of a substance.
- Metals and non-metals show contrasting physical properties due to differences in their structure and bonding.

Property	Metals	Non-metals
1. Lustre	Generally lustrous (shiny).	Generally dull (non-lustrous).
2. Hardness	Generally hard.	Generally soft; brittle if solid.
3. Strength	Strong; can withstand tension and pressure.	Generally weak; break or crumble on applying force.
4. Malleability	Malleable – can be beaten into thin sheets.	Not malleable – brittle if solid.
5. Ductility	Ductile – can be drawn into thin wires.	Not ductile – cannot be drawn into wires.
6. Sonority	Sonorous – produce a ringing sound when struck.	Non-sonorous – do not produce a ringing sound.
7. Conductivity - Heat - Electricity	- Good conductors of heat. - Good conductors of electricity.	- Poor conductors of heat (except graphite). - Poor conductors of electricity (except graphite).
8. Density	Generally high density (exceptions exist).	Generally low density (exceptions exist).
9. Melting Point	Generally high melting points (except: Hg – liquid at room temperature).	Generally low melting points (exceptions exist).
10. Boiling Point	Generally high boiling points.	Generally low boiling points.
11. Physical State at Room Temperature	All metals are solids except mercury (liquid).	May be solids, liquids or gases (e.g., O ₂ , N ₂ , Cl ₂ – gases; Br ₂ – liquid; I ₂ , S – solids).

Important Exceptions (Remember These!)

Mercury (Metal)  Liquid metal at room temperature.	Bromine (Non-metal)  Liquid non-metal at room temperature.	Sodium & Potassium (Metals)  Very soft metals; can be cut with a knife.	Graphite (Non-metal)  Good conductor of electricity.	Iodine (Non-metal)  Lustrous solid (non-metal).	Diamond (Non-metal)  Extremely hard (non-metal).
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Examples in Daily Life

Copper Wire  Ductile and excellent conductor of electricity → used in electrical wiring.	Aluminium Utensil  Malleable, light, good conductor of heat → used for making cooking utensils.	Graphite in Pencil  Graphite conducts electricity and is soft → used in pencil leads and electrodes.	Sulphur Powder  Non-metal, brittle solid, poor conductor → used in making sulphuric acid and matches.	Bromine  Liquid non-metal → used in photography chemicals and flame retardants.
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Why These Properties Matter



Metals are used for:

- Electrical wiring → good conductors and ductile (e.g., Cu, Al)
- Cooking utensils → good conductors of heat (e.g., Al)
- Machinery and tools → strong, hard and durable (e.g., Fe, steel)
- Buildings and transport → strong, malleable and durable (e.g., Fe, Al)



Non-metals are used differently:

- Insulators → electrical (e.g., plastic, rubber) and thermal (wood)
- Chemicals and medicines → react in specific ways (e.g., sulphur)
- Fuels → many non-metals form good fuels (e.g., C in coal, H in H₂)
- Laboratory reagents → used in specific reactions (e.g., acids, halogens)



Exam Tips – Centum Focus

- Learn the physical properties of metals and non-metals side by side.
- Remember key exceptions (table above) – they are favourite exam questions.
- Use keywords: Metals → lustrous, malleable, ductile, good conductors, sonorous. Non-metals → dull, brittle, poor conductors, non-sonorous.
- Diagram-based questions may ask for daily-life uses.
- Distinguish between similar-looking substances using properties (e.g., graphite vs diamond).
- Practice tabular comparison answers for 3, 4 or 5 marks.

Ch 3: Metals and Non-metals

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION C - CHEMICAL PROPERTIES OF METALS: REACTION WITH OXYGEN, WATER, AIR AND STEAM

1. Reaction of metals with oxygen

- Metals react with oxygen from the air to form metal oxides.
- The reaction generally occurs on heating or on prolonged exposure to air.
- Most metal oxides are basic in nature.
- Aluminium oxide (Al_2O_3) and zinc oxide (ZnO) are amphoteric (i.e., they react with both acids and bases).

Metal	Equation (Balanced)	Observation / Colour Change
Magnesium (Mg)	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	Burns with dazzling white light; white ash of MgO forms.
Aluminium (Al)	$4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$	Shining surface turns dull white due to Al_2O_3 coating.
Copper (Cu)	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$	Reddish-brown copper turns black (CuO) on heating.
Iron (Fe)	$3\text{Fe} + 2\text{O}_2 \rightarrow \text{Fe}_3\text{O}_4$	Grey iron turns black (magnetite, Fe_3O_4) on heating.
Lead (Pb)	$2\text{Pb} + \text{O}_2 \rightarrow 2\text{PbO}$	Silvery grey turns yellow (PbO) on heating.

Nature of metal oxides

- Metal oxides (e.g., Na_2O , MgO , CaO , CuO , Fe_2O_3) are generally basic.
- They react with acids to form salt and water.

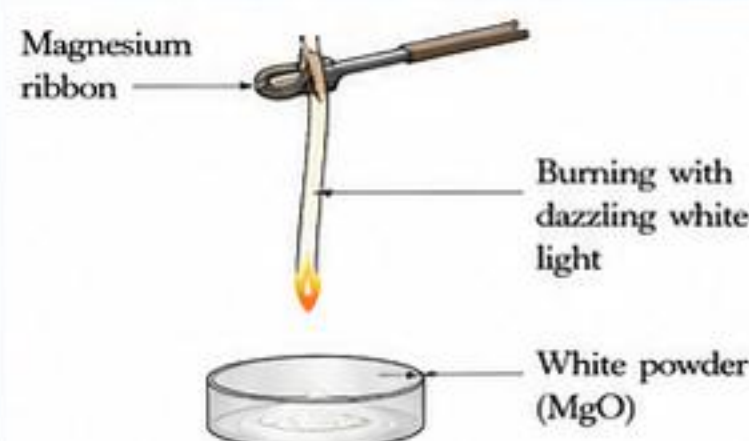
Example: $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$

- Aluminium oxide and zinc oxide are amphoteric.
 - With acids: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
 - With bases: $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ (Similarly for ZnO)

2. Corrosion / Tarnishing (Reaction with air)

- Some metals react slowly with oxygen, moisture and other substances in air and get corroded (e.g., iron) or tarnished (e.g., silver, copper).
- This is an example of a chemical reaction that damages the metal.
- Examples:
 - Iron rusts to form hydrated iron(III) oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).
 - Silver becomes black due to formation of silver sulphide (Ag_2S) when it reacts with traces of H_2S in air.
 - Copper develops a green coating of basic copper carbonate ($\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$).

Burning of magnesium ribbon in air



Equation:
 $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

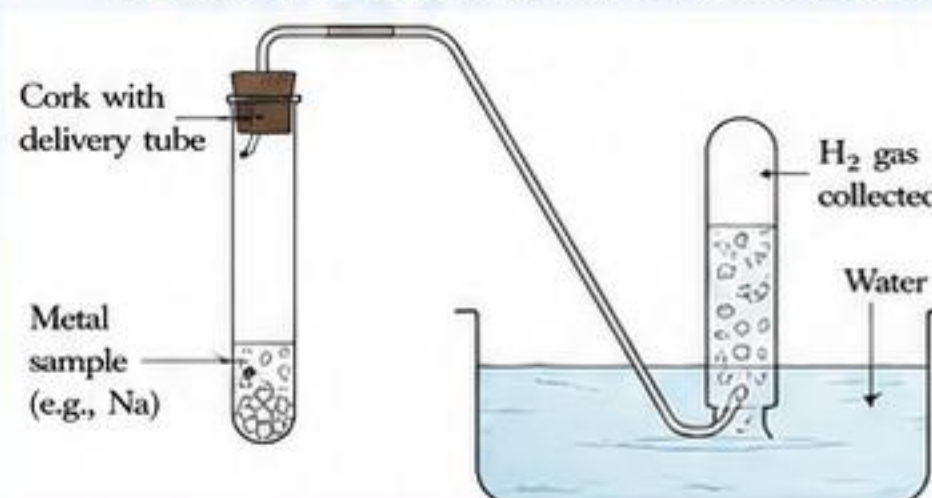
Observation:
 It burns with brilliant white light and forms a white powder of magnesium oxide.

3. Reaction of metals with water

- Reactivity of metals with water depends on the nature of metal and the temperature of water.
- More reactive metals displace hydrogen from water.
- Less reactive metals require steam to react.

Reactivity category	Metals	Water / Steam condition	Products formed	Example equation (Balanced)	Observation
Very reactive	Potassium (K), Sodium (Na)	Cold water	Metal hydroxide + Hydrogen	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	Vigorous reaction; fizzing; heat produced; metal floats and moves.
Reactive	Calcium (Ca)	Cold water	Metal hydroxide + Hydrogen	$\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{H}_2$	Steady reaction; bubbles of H_2 ; solution turns milky.
Moderately reactive	Magnesium (Mg)	Hot water	Magnesium hydroxide + Hydrogen	$\text{Mg} + 2\text{H}_2\text{O} (\text{hot}) \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	Slow reaction with hot water; white precipitate of $\text{Mg}(\text{OH})_2$.
		Steam	Magnesium oxide + Hydrogen	$\text{Mg} + \text{H}_2\text{O} (\text{steam}) \rightarrow \text{MgO} + \text{H}_2$	Burning of Mg with steam; white powder of MgO.
Less reactive	Aluminium (Al), Zinc (Zn), Iron (Fe)	Steam	Metal oxide + Hydrogen	$2\text{Al} + 3\text{H}_2\text{O} (\text{steam}) \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$ $\text{Zn} + \text{H}_2\text{O} (\text{steam}) \rightarrow \text{ZnO} + \text{H}_2$ $3\text{Fe} + 4\text{H}_2\text{O} (\text{steam}) \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$	Slow reaction; metal oxide forms; hydrogen evolves.
Very less reactive	Copper (Cu), Silver (Ag), Gold (Au)	Cold water & Steam	No reaction	—	No bubbles; no change.

Reaction of a metal with water / steam (general setup)



Observation (example with Na):
 Bubbles of hydrogen gas are evolved; heat is released; the solution becomes basic.

Equation (example):
 $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$

Test for hydrogen:
 Burning splint near the mouth of the test tube gives a 'pop' sound.

4. Why sodium and potassium are stored in kerosene?

- Sodium and potassium are very reactive with air and moisture.
- They can catch fire or react violently if exposed to air.
- Therefore, they are stored under kerosene which prevents their with air and moisture.



5. Key Takeaways



- Metals combine with oxygen to form metal oxides (mostly basic).
- Al_2O_3 and ZnO are amphoteric – react with both acids and bases.
- Metals react with water based on their reactivity:
 Very reactive $\rightarrow$ cold water; less reactive $\rightarrow$ steam.
- Copper, silver and gold do not react with water or steam.
- Corrosion and tarnishing occur due to slow reaction with air.



6. Exam Tips – Centum Focus

- Always balance chemical equations in reactions with oxygen, water or steam.
- Remember the reactivity order for water: $\text{K, Na} > \text{Ca} > \text{Mg} > \text{Al, Zn, Fe} > \text{Cu, Ag, Au}$.
- Amphoteric oxides: Al_2O_3 and ZnO .
- Corrosion of iron $\rightarrow$ rust ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).
- State observations (colour change, heat, gas) wherever asked.

Ch 3: Metals and Non-metals

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION D - CHEMICAL PROPERTIES OF METALS AND NON-METALS: REACTION WITH ACIDS, BASES AND OTHER SUBSTANCES

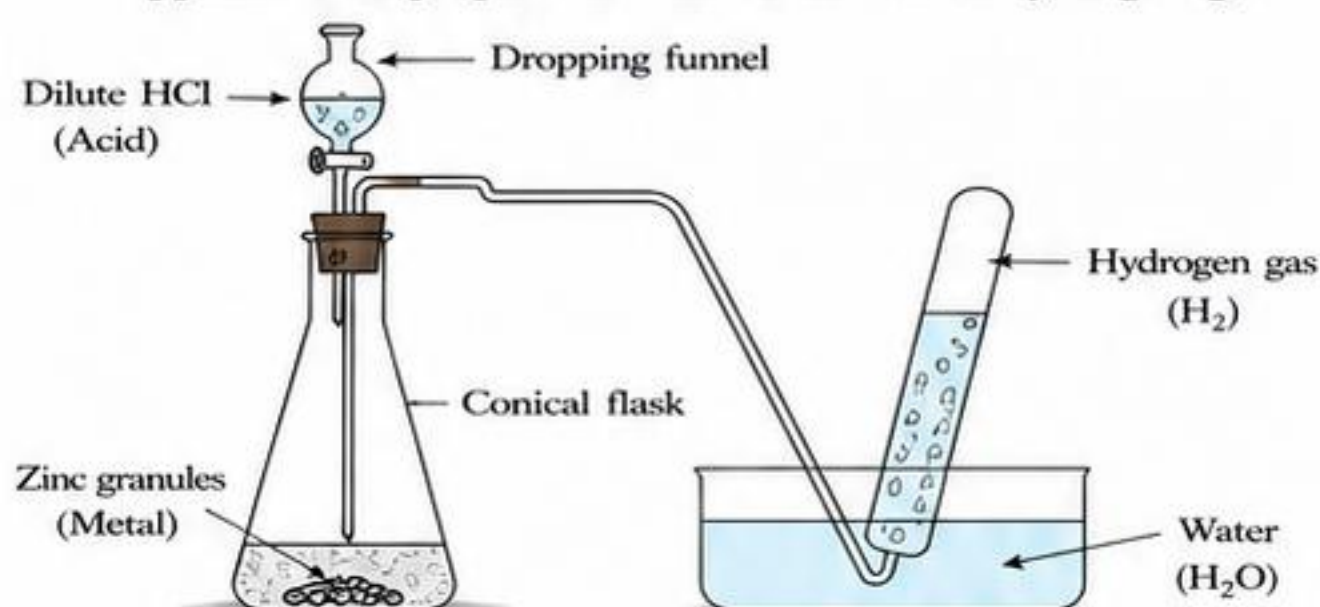
1. Reaction of metals with dilute acids → salt + hydrogen gas

- Most metals above hydrogen in the activity series react with dilute acids to form a salt and liberate hydrogen gas.
- General form:** Metal + Dilute acid → Salt + Hydrogen (H₂)
- Example (Balanced):**

Metal	Acid	Equation (Balanced)
Zinc (Zn)	Dilute HCl	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
Magnesium (Mg)	Dilute H ₂ SO ₄	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$
Iron (Fe)	Dilute HCl	$\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$

- Observations:**
 - Brisk effervescence (bubbles) is seen.
 - The gas produced burns with a 'pop' sound when a burning splint is brought near (test for hydrogen).
 - The metal gradually dissolves.

Apparatus for preparation and collection of hydrogen gas

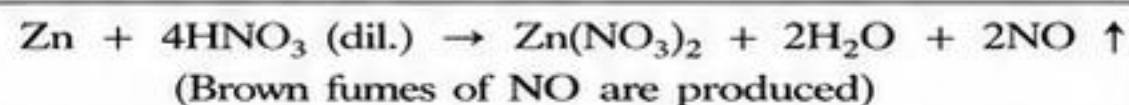


Hydrogen is collected over water by downward displacement of water. It is almost insoluble in water.

Test for hydrogen: Bring a burning splint near the mouth of the test tube → gas burns with a 'pop' sound.

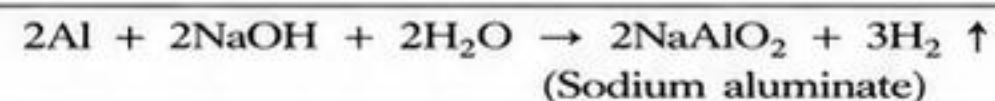
2. Important exception – Nitric acid

- Nitric acid (HNO₃) is a strong oxidising agent.
- It usually does not liberate hydrogen in the normal way.
- Instead, it is reduced to oxides of nitrogen.
- Example (with dilute HNO₃):**



3a. Aluminium with strong base (NaOH)

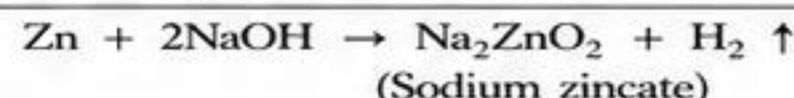
- Aluminium dissolves in aqueous NaOH to form sodium aluminate and hydrogen gas.
- Equation (Balanced):**



- Observations:**
 - Metal dissolves.
 - Brisk effervescence of hydrogen.

3b. Zinc with strong base (NaOH)

- Zinc dissolves in aqueous NaOH to form sodium zincate and liberate hydrogen gas.
- Equation (Balanced):**



- Observations:**
 - Zinc granules dissolve.
 - Effervescence of hydrogen.

4. Reaction of non-metals with oxygen

- Non-metals generally react with oxygen to form oxides.
- Most non-metal oxides are acidic oxides; some are neutral.
- Examples (Balanced):**

Non-metal	Reaction with oxygen	Type of oxide
Carbon (C)	$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$	Acidic oxide
Sulphur (S)	$\text{S} + \text{O}_2 \rightarrow \text{SO}_2$	Acidic oxide
Phosphorus (P)	$4\text{P} + 5\text{O}_2 \rightarrow 2\text{P}_2\text{O}_5$	Acidic oxide
Nitrogen (N)	$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$	Neutral oxide
Silicon (Si)	$\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$	Acidic oxide

5. Reaction of non-metals with other substances

5a. With hydrogen

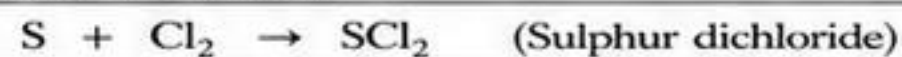
- Some non-metals combine with hydrogen to form covalent compounds.
- Example (Balanced):**



- Other examples: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (Ammonia); $\text{S} + \text{H}_2 \rightarrow \text{H}_2\text{S}$

5b. With chlorine

- Many non-metals react with chlorine to form chlorides.
- Examples (Balanced):**



6. Summary comparison

Property / Reaction	Metals	Non-metals
With dilute acids	React to form salt + H ₂ gas (except with oxidising acids like HNO ₃)	Usually do not react with dilute acids
With strong bases	Amphoteric metals (Al, Zn) react to form complex salt + H ₂	Generally do not react
With oxygen	Form basic or amphoteric oxides	Form acidic or neutral oxides
With hydrogen	Most metals do not react	Many non-metals react to form covalent hydrides
With chlorine	Form ionic chlorides (e.g., NaCl)	Form covalent chlorides (e.g., PCl ₃ , SCl ₂)

3. Reaction of amphoteric metals with strong bases

- Some metals (amphoteric) react with strong bases (e.g., NaOH) to form complex salts (aluminates or zincates) and liberate hydrogen gas.
- These metals react with both acids and bases.
- Common amphoteric metals: Aluminium (Al), Zinc (Zn), Tin (Sn), Lead (Pb) (weakly).



7. Exam Tips – Centum Focus

- Metals above hydrogen react with dilute acids to give H₂ gas.
- Always write balanced equations.
- Remember: H₂ gas test → 'Pop' sound with burning splint.
- Nitric acid is an oxidising acid – it usually does not give H₂.
- Amphoteric metals react with NaOH to give complex salts and H₂.
- Non-metals form oxides with oxygen and covalent compounds with H₂/Cl₂.

Ch 3: Metals and Non-metals

Part 1 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION E - REACTIVITY SERIES AND DISPLACEMENT REACTIONS

1. What is the reactivity series?

- The reactivity series is a list of metals arranged in order of their reactivity.
- Metals at the top are most reactive; those at the bottom are least reactive.
- Reactivity here means the tendency of a metal atom to lose electrons and form positive ions (cations).
- More reactive metals lose electrons more easily.

2. Full reactivity series of metals

Most reactive (lose electrons easily)		Order	Metal	Symbol
		1	Potassium	K
		2	Sodium	Na
		3	Calcium	Ca
		4	Magnesium	Mg
		5	Aluminium	Al
		6	Zinc	Zn
		7	Iron	Fe
Least reactive (lose electrons with difficulty)		8	Lead	Pb
		—	Hydrogen (reference)	H
		9	Copper	Cu
		10	Mercury	Hg
		11	Silver	Ag
		12	Gold	Au

3. What does the series mean?

- Metals higher in the series have a greater tendency to lose electrons.
- They react more vigorously with water, acids and other substances.
- Metals lower in the series have less tendency to lose electrons.
- Hydrogen is taken as the reference point.

4. Why are metals higher in the series more reactive?

- Their atoms have weaker hold on their valence electrons.
- They have lower ionisation energy.
- They more easily lose electrons to form positive ions.
- Hence, they react readily and displace less reactive metals from their compounds.

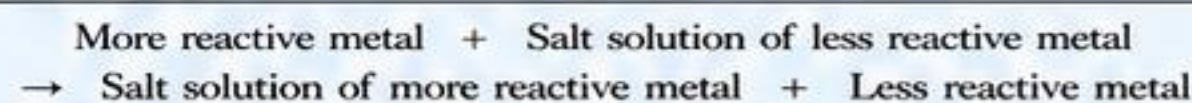
5. Rule for displacement reactions



In words:

A metal higher in the reactivity series displaces (takes out) a metal lower in the series from its salt solution.

General form:



6. Displacement reactions – examples with observations

No.	Reaction	What you see (Observation)	Colour change / Result	Inference
1	$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$	- Reddish-brown coating of copper forms on the zinc strip. - Blue colour of CuSO_4 solution fades.	Blue solution (CuSO_4) → Colourless solution (ZnSO_4) + Reddish-brown Cu on zinc	Zn is more reactive than Cu.
2	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	- Reddish-brown copper deposits on iron. - Blue colour of solution turns green.	Blue solution (CuSO_4) → Green solution (FeSO_4) + Reddish-brown Cu on iron	Fe is more reactive than Cu.
3	$\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu(NO}_3)_2 + 2\text{Ag}$	- Shiny silver crystals deposit on copper. - Solution remains colourless but becomes light blue.	Colourless solution (AgNO_3) → Light blue solution ($\text{Cu(NO}_3)_2$) + Shiny silver Ag on copper	Cu is more reactive than Ag.
4	$\text{Cu} + \text{FeSO}_4 \rightarrow$ No reaction	- No change on copper strip. - Solution remains green.	Green solution (FeSO_4) → Green solution (FeSO_4) (no change)	Cu is less reactive than Fe.



7. Key points to remember

- A metal can displace another metal only if it is above it in the reactivity series.
- If the metal is below, no reaction occurs.
- Hydrogen line (H): Metals above H displace hydrogen from acids; metals below H do not.
 - Example: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$ (reacts)
 - $\text{Cu} + 2\text{HCl} \rightarrow$ No reaction
- More reactive metals can also reduce metal oxides of less reactive metals (thermite type reactions).

8. Reactivity and extraction (very brief)

Position in reactivity series	Method of extraction
Very high (K, Na, Ca, Mg, Al)	Electrolysis of molten compounds (not by carbon reduction)
Medium (Zn, Fe, Pb)	Reduction by carbon/CO in a furnace (smelting)
Low (Cu, Hg, Ag, Au)	Found in native state or extracted by simple methods (e.g., roasting, leaching)

9. Colour observations quick guide

- CuSO_4 solution = Blue
- FeSO_4 solution = Green (pale green)
- ZnSO_4 solution = Colourless
- AgNO_3 solution = Colourless
- $\text{Cu(NO}_3)_2$ solution = Light blue
- Copper metal = Reddish-brown deposit
- Silver metal = Shiny grey crystals



10. Memory tip

Top to Bottom = Strong to Weak!

Top metals are "eager to lose" electrons.
Bottom metals "hold on" to electrons tightly.

K Na Ca Mg Al | Zn Fe Pb | H | Cu Hg Ag Au

"Keep Naughty Cats Making Aluminium; Zinc Fights Poor Hydrogen; Copper Hugs Auntie Gold."



11. Exam Tips – Writing displacement reactions

- Always write the more reactive metal first.
- Write the correct formula of the salt solution.
- Balance the equation (check ions and atoms).
- Mention state symbols where asked: (s), (aq).
- State observations: colour change, metal deposit, bubbles.
- Conclude by comparing the reactivity of the metals.

12. Centum Focus

- ★ Learn the full reactivity series in order.
- ★ Understand why hydrogen is the reference.
- ★ Practice at least 5 displacement reactions.
- ★ Be ready to predict "will react / will not react" quickly.
- ★ Link series with extraction methods.
- ★ Draw neat diagrams of colour change setups.

Ch 3: Metals and Non-metals

Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION F - ELECTRONIC TENDENCY, ION FORMATION AND IONIC BONDING

1. Valence electrons (brief)

- Valence electrons are the electrons present in the outermost shell of an atom.
- These electrons take part in chemical reactions.

2. Electronic tendency and ion formation

- Atoms are most stable when their outermost shell is completely filled (octet) or has 2 electrons (duplet for first shell).
- Metals have 1, 2 or 3 valence electrons. They tend to lose these electrons to achieve a stable configuration of the nearest noble gas.
- Non-metals have 4 to 7 valence electrons. They tend to gain electrons to complete the octet.
- On losing electrons, atoms form positively charged ions called cations. On gaining electrons, atoms form negatively charged ions called anions.

3. Why this happens?

- Atoms with incomplete valence shells are less stable and more reactive.
- By losing or gaining electrons, they achieve a stable electronic configuration like that of noble gases (duet/octet).
- This stability is the driving force for ion formation.

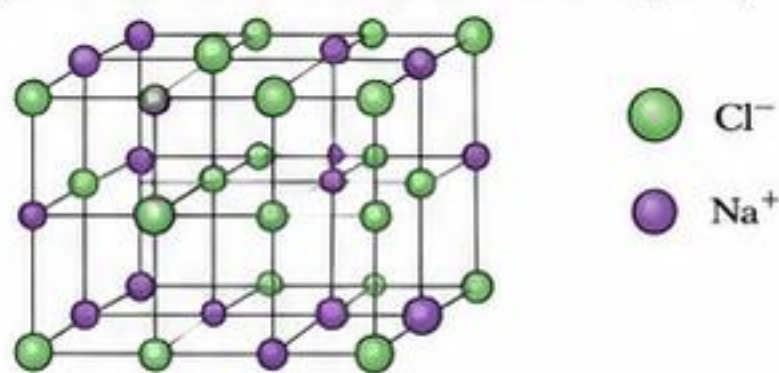
4. Electron transfer in ionic compounds

- Ionic compounds are formed by complete transfer of electrons from a metal atom to a non-metal atom.
- The oppositely charged ions attract each other by electrostatic force to form an ionic compound.

6. Ionic lattice (arrangement of ions)

- Ions in an ionic compound are arranged in a regular 3-D pattern called an ionic lattice.
- Each cation is surrounded by a fixed number of anions and vice-versa.
- Strong electrostatic attraction holds the lattice together.

Example: Ionic lattice of Sodium chloride (NaCl)



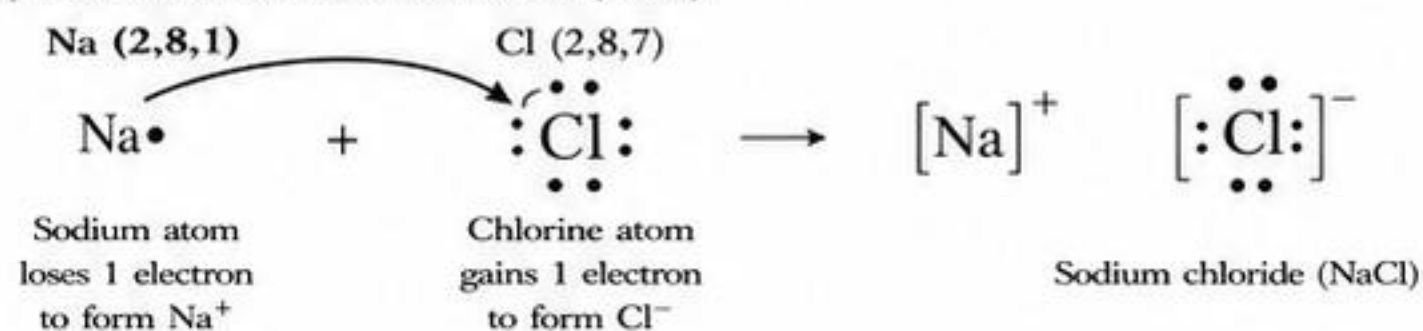
Each Na⁺ is surrounded by 6 Cl⁻ ions and each Cl⁻ is surrounded by 6 Na⁺ ions.

8. Comparison: Ionic compounds vs Covalent substances

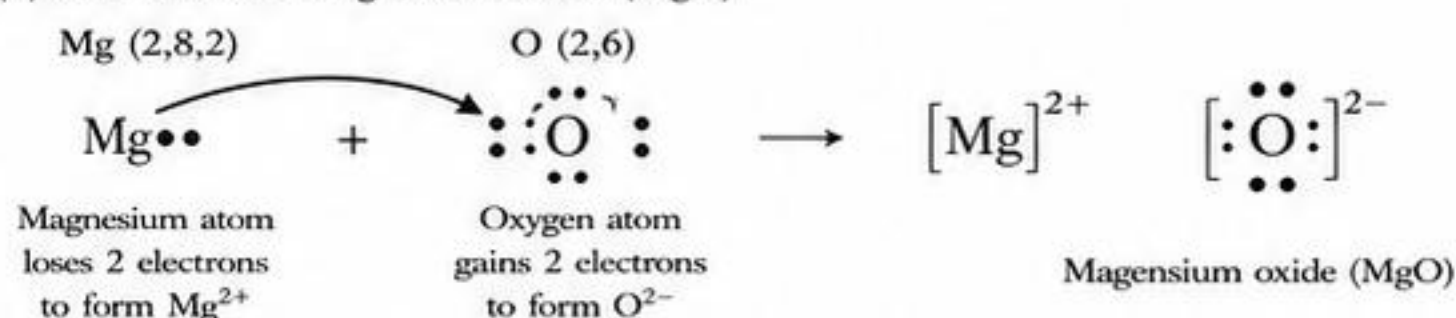
Property	Ionic Compounds	Covalent Substances
Constituent particles	Ions (cations and anions)	Molecules (neutral)
Bond type	Ionic bond (electrostatic attraction between ions)	Covalent bond (sharing of electrons)
Physical state	Generally solids	Solids, liquids or gases
Melting and boiling points	High	Low to moderate
Hardness	Hard but brittle	Usually soft
Solubility in water	Generally soluble	Generally insoluble (except some molecules)
Electrical conductivity	Conduct in molten state and aqueous solution; not in solid	Do not conduct in any state (except graphite)

5. Formation of ionic compounds (Electron transfer and electron-dot diagrams)

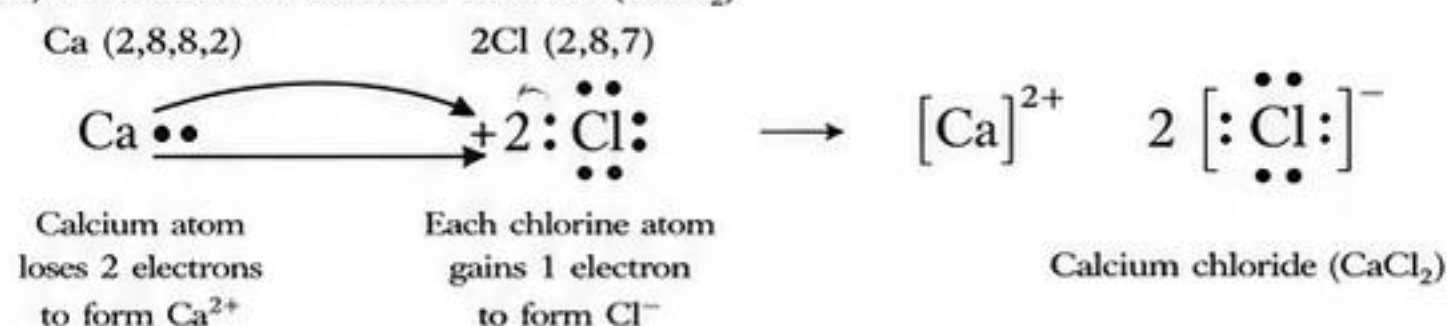
(i) Formation of Sodium chloride (NaCl)



(ii) Formation of Magnesium oxide (MgO)



(iii) Formation of Calcium chloride (CaCl₂)



7. Properties of ionic compounds (and reasons)

Property	Observation	Reason
(i) Hard and brittle	Ionic compounds are hard but brittle (they break on hammering).	Ions are held strongly in a rigid lattice. When force is applied, layers of ions shift. Like charges come next to each other and repel strongly, causing the crystal to break.
(ii) High melting and boiling points	Ionic compounds have very high melting and boiling points.	A large amount of energy is required to break the strong electrostatic forces between oppositely charged ions in the lattice.
(iii) Generally soluble in water	Most ionic compounds dissolve in water.	Water is a polar solvent. It attracts and separates the ions (hydration), overcoming the lattice forces and dissolving the compound.
(iv) Conduct electricity in molten or aqueous state	Molten ionic compounds and their aqueous solutions conduct electricity.	In molten state or solution, ions are free to move and carry electric current.
(v) Do not conduct electricity in solid state	Solid ionic compounds do not conduct electricity.	In solid state, ions are fixed in the lattice and cannot move. Hence, no current flows.

9. Exam Tip (Centum Focus)



- Remember: Metals lose electrons → cations; Non-metals gain electrons → anions.
- Ionic compounds = metal + non-metal → electron transfer → ionic bond → ionic lattice.
- Link properties with reasons in exam answers for full marks.



10. Quick Exam Box

Q. Why do ionic compounds not conduct electricity in solid state?
Ans. In solid ionic compounds, ions are arranged in a rigid lattice. They are held at fixed positions and cannot move. Since electric current is due to the movement of charged particles, no current flows in the solid state. However, when molten or dissolved in water, ions are free to move and conduct electricity.

Ch 3: Metals and Non-metals

Part 2 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION G - OCCURRENCE OF METALS, MINERALS, ORES AND METALLURGY OVERVIEW

1. Key definitions

- Mineral:** A naturally occurring substance in the Earth's crust having a definite chemical composition and crystalline structure.
Example: Quartz (SiO_2), Calcite (CaCO_3).
- Ore:** A mineral (or a combination of minerals) from which a metal can be extracted profitably.
- Gangue (Matrix):** The unwanted earthy or rocky impurities that are mixed with the ore.
- Flux:** A substance added during extraction to combine with gangue and form a fusible slag.
Examples: CaO (Quicklime), SiO_2 (Sandstone), CaCO_3 (Limestone).
- Slag:** The fusible product formed by the reaction of flux with gangue. It is lighter than the molten metal and floats on top, making it easy to separate.

2. Mineral vs Ore

Aspect	Mineral	Ore
Definition	Naturally occurring substance with definite composition and structure.	Mineral (or group of minerals) from which a metal can be extracted economically.
Occurrence	May be abundant but metal content may be low.	Contains sufficient metal in concentrated form.
Importance	Scientific / geological value.	Economic value (source of metal).
Examples	Quartz (SiO_2), Feldspar (KAlSi_3O_8), Calcite (CaCO_3)	Haematite (Fe_2O_3 – iron ore), Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ – aluminium ore)

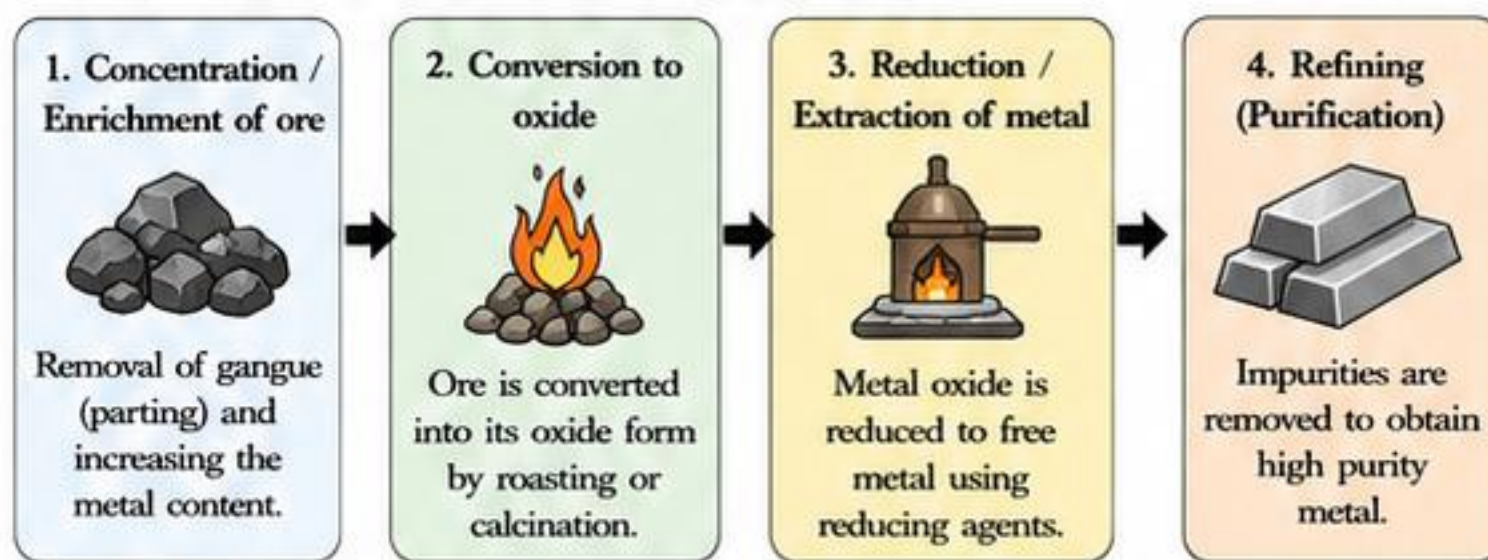
3. Why only some minerals are ores?

- Metals are rarely found in the free (native) state; they occur as compounds (minerals).
- For a mineral to be an ore, it must:
 - Contain the metal in sufficient amount (high % of metal).
 - Be easily and economically extractable.
 - Occur in large enough quantity.
- Example:** Bauxite is an ore of aluminium; but clay (Al_2O_3 with impurities in very low amount) is not an ore.

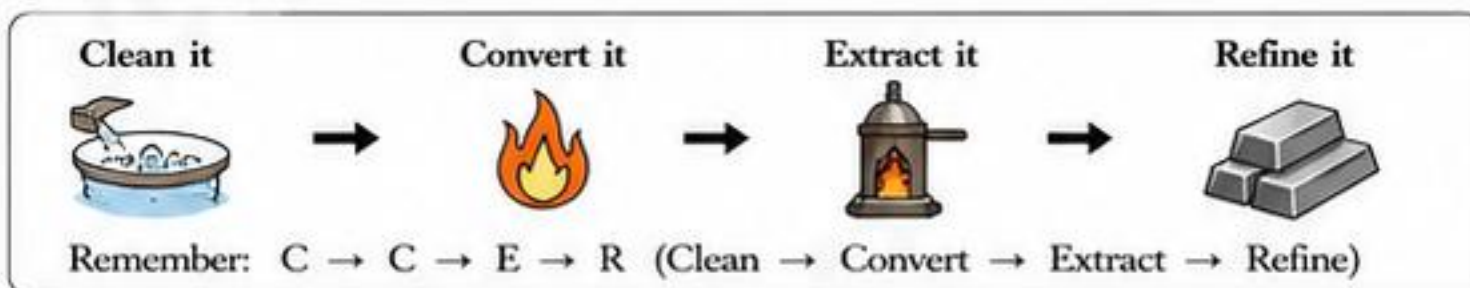
4. Important ores and the metals obtained

Ore (Name)	Formula	Metal obtained	Main source / Occurrence
Bauxite (Hydrated alumina)	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$	Aluminium (Al)	Laterites; rich in $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$, clay and silica.
Haematite (Red iron ore)	Fe_2O_3	Iron (Fe)	Widely occurs in sedimentary rocks.
Magnetite (Magnetic iron ore)	Fe_3O_4	Iron (Fe)	Igneous and metamorphic rocks; magnetic.
Cinnabar (Mercury sulphide)	HgS	Mercury (Hg)	Occurs in volcanic regions; converted to Hg on roasting.
Calamine (Zinc carbonate)	ZnCO_3	Zinc (Zn)	Occurs in sedimentary rocks of zinc deposits.
Zinc blende (Zinc sulphide)	ZnS	Zinc (Zn)	Most important zinc ore; widespread.
Galena (Lead sulphide)	PbS	Lead (Pb)	Common lead ore; occurs in rock veins.

5. Overview of metallurgy → Major steps



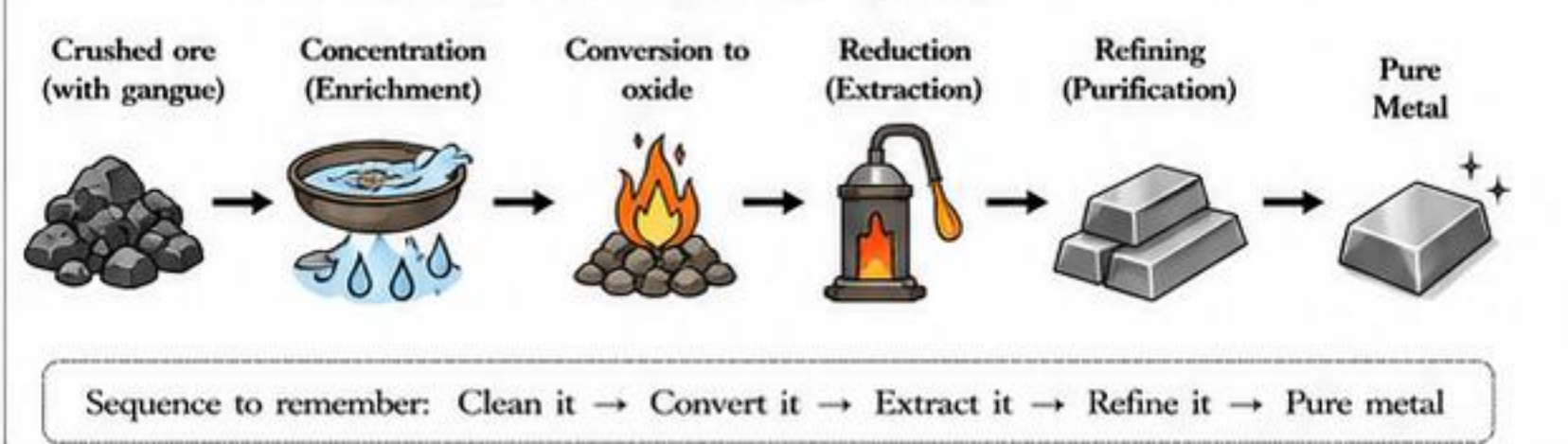
Easy memory aid:



6. Methods of concentration (enrichment) of ores

Method	Principle	What it does (one line)	Example ores
Hydraulic washing	Based on difference in density; lighter gangue is washed away by running water.	Removes lighter earthy impurities by washing.	Bauxite, Tin ore (cassiterite), Gold (placer deposits)
Magnetic separation	Based on difference in magnetic properties; magnetic minerals are attracted by a magnet.	Separates magnetic particles from non-magnetic gangue.	Magnetite (Fe_3O_4) from silica, Haematite (weakly magnetic)
Froth flotation	Based on difference in wettability; ore particles stick to froth and gangue remains in water.	Useful for sulphide ores; concentrates the ore in froth.	Zinc blende (ZnS), Galena (PbS), Copper pyrites (CuFeS_2)
Leaching	Based on difference in solubility; ore is dissolved in suitable reagent, impurities remain undissolved.	Extracts soluble ore selectively by chemical solution.	Bauxite (Bayer's process), Uranium ores, Low grade Ni ores

7. Metallurgy flow diagram (at a glance)



8. Exam Tips – Centum Focus

- Know definitions: mineral, ore, gangue, flux, slag.
- Learn the difference between mineral and ore with examples.
- Memorise important ores with formula and metal obtained.
- Remember concentration methods and their principles.
- Understand the 4 major steps of metallurgy.
- Practice flowchart-based and comparison questions.

Ch 3: Metals and Non-metals

Part 2 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION H - EXTRACTION OF METALS OF LOW AND MEDIUM REACTIVITY

1. Occurrence of low-reactivity metals

- Very low-reactivity metals like gold (Au), silver (Ag), platinum (Pt) and sometimes copper (Cu) occur in native (free) state in nature.
- These metals are not combined chemically in the earth's crust because they do not react with oxygen, water or other substances under normal conditions.
- Hence, they can be obtained by simple physical methods.
- Example:** Gold is found in river sand and can be collected by washing (gravity separation).

2. Extraction of low-reactivity metals

- If present as **native metals** – obtained by physical methods like washing, crushing, sieving, magnetic separation, etc.
- If present as compounds (oxides, sulphides, chlorides etc.) – the ores are first concentrated, then the compounds are converted into metals by heating.
- Oxides of these metals are decomposed by strong heating.

General forms:

- Metal oxide $\longrightarrow$ Metal + Oxygen
- Metal sulphide $\longrightarrow$ Metal + Sulphur dioxide (or H_2S with steam)

Examples

- $2\text{Ag}_2\text{O} \xrightarrow{\Delta} 4\text{Ag} + \text{O}_2$ (Silver oxide)
- $2\text{Cu}_2\text{O} \xrightarrow{\Delta} 4\text{Cu} + \text{O}_2$ (Copper(I) oxide)
- $2\text{PtO}_2 \xrightarrow{\Delta} 2\text{Pt} + 2\text{O}_2$ (Platinum oxide)

In many cases, roasting is done first to convert sulphides to oxides and then the oxide is heated to obtain the metal.

3. Extraction of medium-reactivity metals – Overview

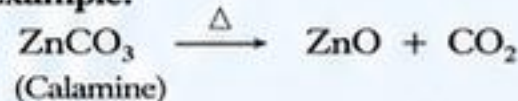
- Metals like Zn, Fe, Pb, Sn, Ni, etc., are more reactive than Cu but less reactive than Na, Mg, Al, etc.
- They do not occur in native state.
- Their ores are generally found as carbonates, sulphides or oxides.
- Extraction involves three main steps:
 - Concentration of ore (removal of gangue)
 - Conversion of ore to metal oxide (calcination or roasting)
 - Reduction of metal oxide to metal

4. Calcination and roasting

Calcination (for carbonates)

- Strong heating of carbonate ores in limited or no air.
- Converts carbonate to oxide by driving off CO_2 .

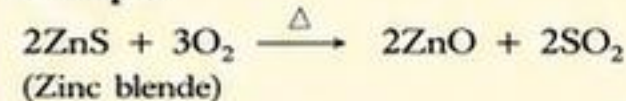
Example:



Roasting (for sulphide ores)

- Strong heating of sulphide ores in excess of air (oxygen).
- Converts sulphide to oxide by removing sulphur as SO_2 .

Example:



Note: Choice of method depends on the nature of the ore.

5. Why ores are converted into oxides before reduction?

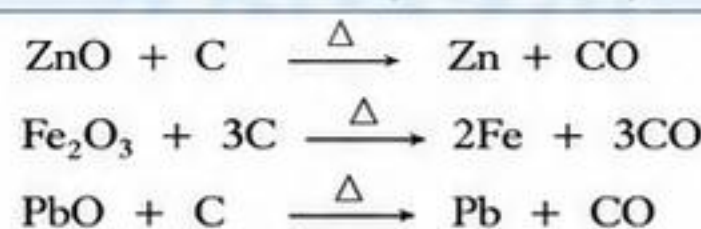
- Oxides are easier to reduce to metals than carbonates or sulphides.
- Carbonates contain CO_2 and sulphides contain sulphur which must be removed first.
- Oxides are more stable, easier to handle and allow efficient reduction.

6. Reduction of metal oxides

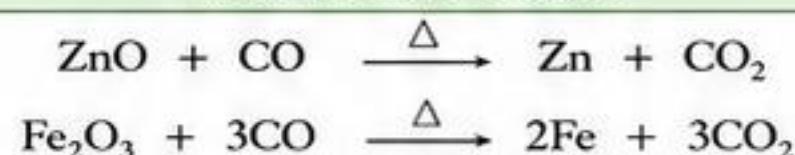
- Metal oxides are reduced to metals using suitable reducing agents.
- Common reducing agents: Carbon (C) or Carbon monoxide (CO).
- High temperature is required.

General reactions:

With carbon (coke/charcoal)



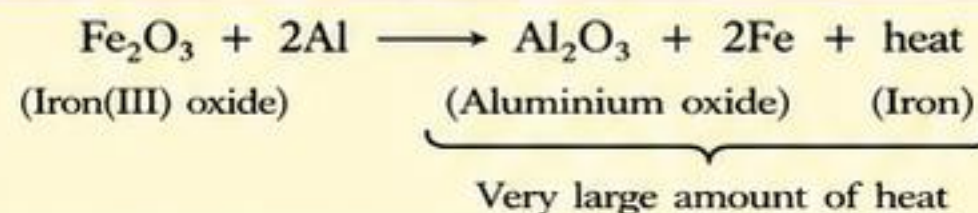
With carbon monoxide



These reactions take place in furnaces such as blast furnace or reverberatory furnace.

7. Thermite reaction (aluminothermic reduction)

- Some metal oxides are very stable and cannot be reduced easily by carbon or CO (e.g., Cr_2O_3 , MnO_2).
- They are reduced by highly reactive metals like aluminium.
- This is called thermite reaction.
- Example** (thermite reaction for iron):



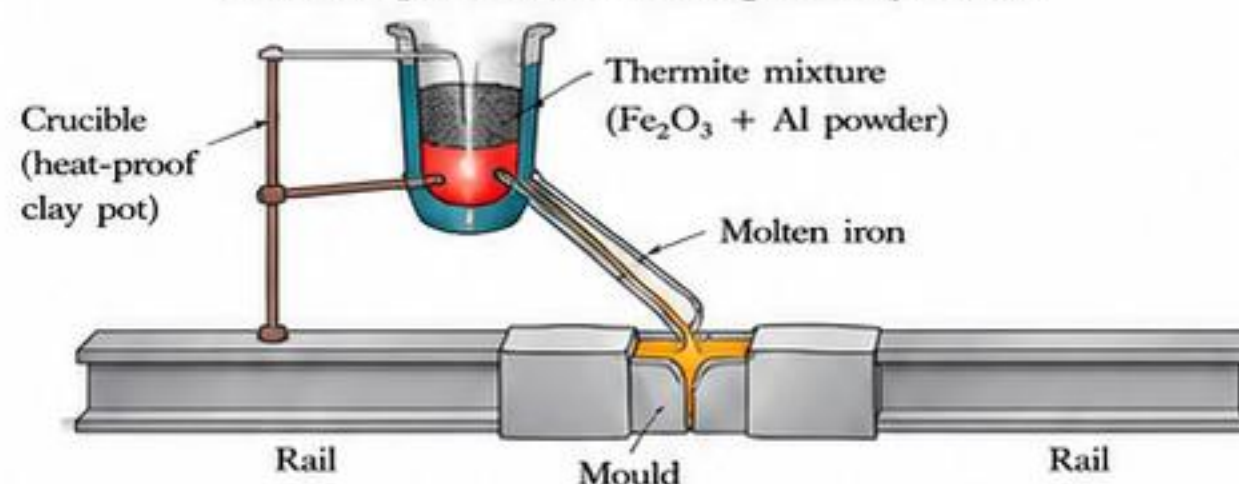
Why is it highly exothermic?

- Aluminium has a very strong affinity for oxygen and forms Al_2O_3 , which is very stable.
- A large amount of energy is released in the formation of Al_2O_3 .

Uses:

- Employed in welding and joining railway tracks. Molten iron produced in situ flows into the gap between rail ends and solidifies, joining them strongly.

Thermite process for welding railway tracks



Molten iron fills the gap and solidifies to join the rails.



9. Exam Tips – Centum Focus

- Remember native metals: Au, Ag, Pt (always); Cu (sometimes).
- Calcination: carbonate $\rightarrow$ oxide + CO_2 (limited/no air).
- Roasting: sulphide $\rightarrow$ oxide + SO_2 (excess air).
- Oxides are reduced by C or CO: write balanced equations.
- Thermite: $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe} + \text{heat}$ (very exothermic).
- Know the use: Thermite reaction is used in welding railway tracks.

Ch 3: Metals and Non-metals

Part 2 of 2 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION I - EXTRACTION OF HIGHLY REACTIVE METALS AND ELECTROLYTIC REFINING

1. Why some metals cannot be obtained by reduction with carbon

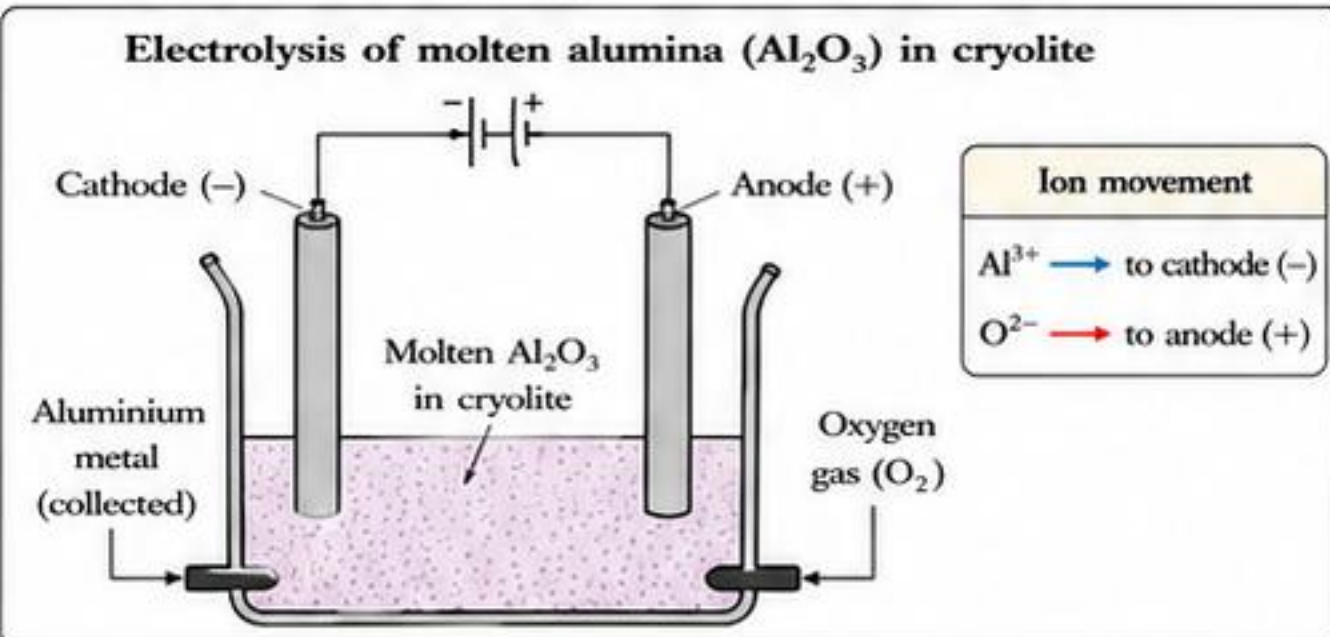
- Highly reactive metals like potassium (K), sodium (Na), calcium (Ca), magnesium (Mg) and aluminium (Al) are at the top of the reactivity series.
- These metals have a greater affinity for oxygen than carbon.
- Their oxides are very stable and cannot be reduced by carbon or by carbon monoxide.
- Hence, they cannot be obtained by the usual method of reduction with carbon.

2. Extraction by electrolysis of molten salts or fused compounds

- These metals are extracted by electrolysis of their molten salts (halides, chlorides) or fused oxides.
- When electric current is passed through the molten compound:
 - Cations (positive ions) move towards the cathode (-) and get reduced to form the metal.
 - Anions (negative ions) move towards the anode (+) and get oxidised to give gases.
- This method is suitable because the metals are in the liquid state and ions can move freely.

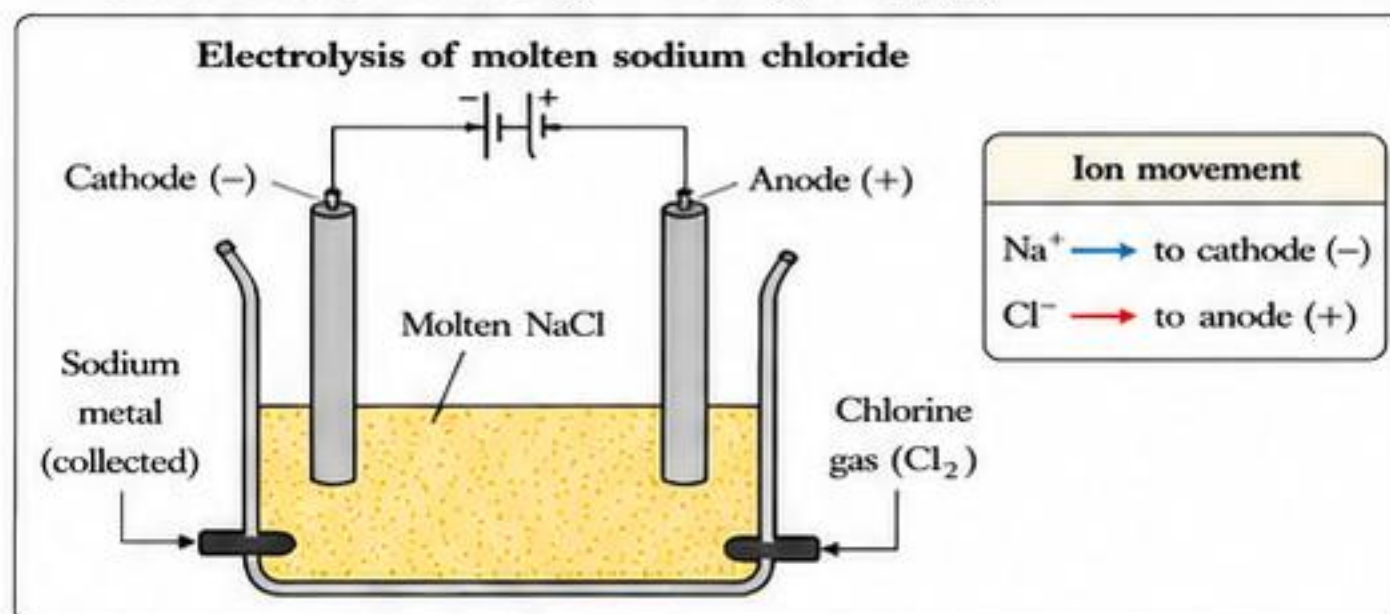
4. Electrolysis of molten aluminium oxide (alumina, Al_2O_3)

- Alumina is first purified and then dissolved in molten cryolite (Na_3AlF_6) to lower its melting point (about 950°C).
- Products: Aluminium metal at cathode and oxygen gas at anode.
- Simplified reactions:
 - At cathode (-): $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ (aluminium metal)
 - At anode (+): $2\text{O}^{2-} \rightarrow \text{O}_2 (\text{g}) + 4\text{e}^-$
- Overall reaction: $2\text{Al}_2\text{O}_3 (\text{l}) \rightarrow 4\text{Al} (\text{l}) + 3\text{O}_2 (\text{g})$



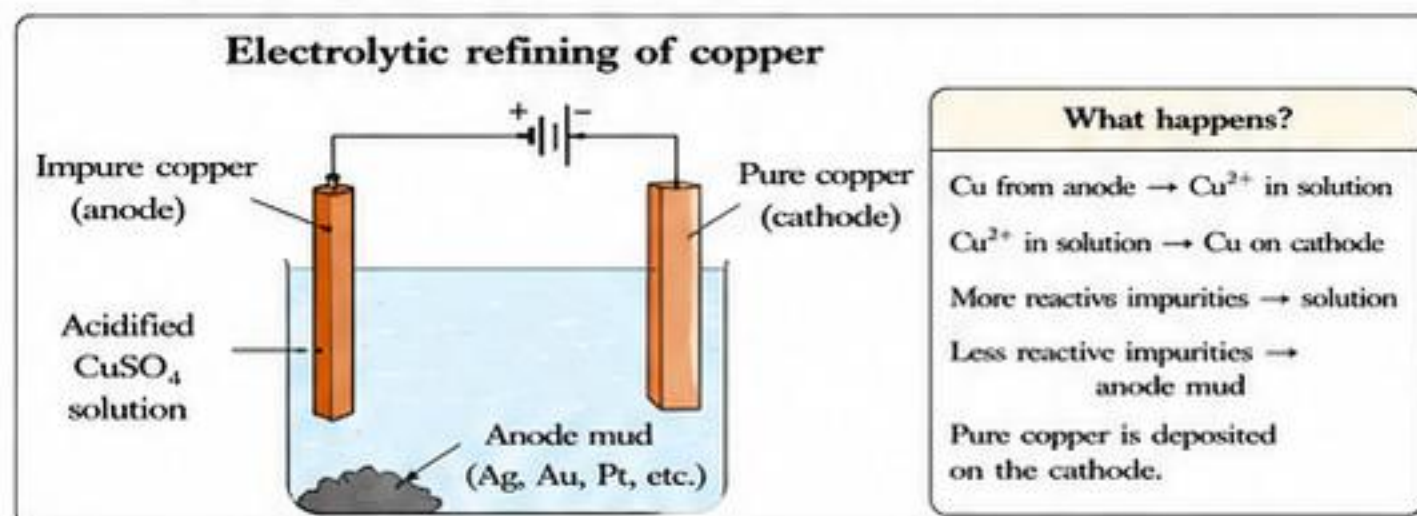
3. Electrolysis of molten sodium chloride (NaCl)

- Sodium chloride is in molten state (about 801°C).
- Products: Sodium metal at cathode and chlorine gas at anode.
- Simplified reactions:
 - At cathode (-): $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ (sodium metal)
 - At anode (+): $2\text{Cl}^- \rightarrow \text{Cl}_2 (\text{g}) + 2\text{e}^-$
- Overall reaction: $2\text{NaCl} (\text{l}) \rightarrow 2\text{Na} (\text{l}) + \text{Cl}_2 (\text{g})$



5. Electrolytic refining of copper

- Impure copper obtained after smelting contains impurities like Fe, Zn, Ag, Au, etc.
- It is purified by electrolytic refining.
- Arrangement:
 - Anode (+): A thick slab of impure copper.
 - Cathode (-): A thin sheet of pure copper.
 - Electrolyte: Acidified copper sulphate solution.
- When current is passed:
 - Copper from the anode goes into the solution as Cu^{2+} ions.
 - Cu^{2+} ions are deposited on the cathode as pure copper.
 - More reactive impurities (like Fe, Zn) go into the solution.
 - Less reactive impurities (like Ag, Au, Pt) do not dissolve and collect below the anode as anode mud.



Aspect	Extraction of Highly Reactive Metals (By Electrolysis)	Electrolytic Refining of Copper
Metals involved	K, Na, Ca, Mg, Al and other highly reactive metals	Copper (from impure to pure)
Why electrolysis?	Oxides are too stable; cannot be reduced by carbon	To remove impurities and obtain pure copper
Electrolyte	Molten salts (e.g., NaCl) or fused oxides (e.g., Al_2O_3 in cryolite)	Acidified CuSO_4 solution
Cathode (-)	Receptacle/plate where metal is deposited	Thin sheet of pure copper
Anode (+)	Inert anode (carbon/graphite)	Thick slab of impure copper
Products at cathode	Metal (e.g., Na, Al)	Pure copper
Products at anode	Gases (e.g., Cl_2 from Cl^- , O_2 from O^{2-})	Cu^{2+} ions formed; impurities dissolve or fall as mud
Special point	Ions move freely in molten state	Anode mud collects Ag, Au, Pt, etc.

6. Key Takeaways



- Highly reactive metals are extracted by electrolysis of molten salts or fused oxides.
 - Cations go to cathode (reduction); anions go to anode (oxidation).
- Electrolytic refining gives pure metal: anode = impure, cathode = pure.
- Impurities more reactive $\rightarrow$ solution; less reactive $\rightarrow$ anode mud.



7. Exam Tips – Centum Focus

- Learn the reason why K, Na, Ca, Mg and Al are not extracted by carbon.
- Always write cation $\rightarrow$ cathode and anion $\rightarrow$ anode.
- For reactions, write half-reactions at cathode and anode separately.
- For copper refining, remember: anode (impure), cathode (pure), electrolyte (acidified CuSO_4), anode mud.
- Label diagrams neatly: anode (+), cathode (-), electrolyte, products.

Ch 3: Metals and Non-metals

Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION J - PART 1 MASTER RECAP, MUST-MEMORISE CONCEPTS, NCERT TRAPS AND HIGH-SCORE SHEET

1. Compact Definitions – Must Memorise

Term	Definition
Metal	An element that is lustrous, malleable, ductile, sonorous, a good conductor of heat and electricity and tends to lose electrons to form positive ions.
Non-metal	An element that is dull, brittle (if solid), non-malleable, non-ductile, poor conductor of heat and electricity and tends to gain electrons to form negative ions.
Metalloid	An element that shows properties intermediate between metals and non-metals.
Mineral	A naturally occurring substance in the earth's crust having a definite chemical composition.
Ore	A mineral from which a metal can be extracted profitably.
Gangue	Impurities (earth/sand/rock) associated with an ore.
Ionic bond	The electrostatic attraction between oppositely charged ions formed by complete transfer of electrons.
Reactivity series	A series of metals arranged in order of their reactivity.
Displacement reaction	A reaction in which a more reactive element displaces a less reactive element from its compound.
Calcination	Heating of an ore strongly below its melting point in the absence or limited supply of air.
Roasting	Heating of a sulphide ore in excess air/oxygen.

2. Key Equations – Reactions, Observations & Inferences

S. No.	Equation	Observation	Inference
1.	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$	Brisk effervescence, 'pop' with burning splint	Hydrogen gas evolved
2.	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$	Vigorous effervescence	Hydrogen gas evolved
3.	$\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$	Moderate effervescence	Hydrogen gas evolved
4.	$\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$	Black solid dissolves, solution turns blue-green	CuO (basic oxide) reacts with acid
5.	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	No visible change, solution warm	Neutralisation reaction
6.	$\text{H}_2\text{SO}_4 + 2\text{KOH} \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$	No visible change, solution warm	Neutralisation reaction
7.	$\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$	Brisk effervescence, CO_2 turns limewater milky	CO_2 gas evolved
8.	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	Brisk effervescence, CO_2 turns limewater milky	CO_2 gas evolved
9.	$2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$	White ppt (CaCO_3) turns milky	CO_2 gas absorbed by limewater
10.	$\text{Metal} + \text{Acid} \rightarrow \text{Salt} + \text{H}_2$	Effervescence, pop sound	Displacement reaction; H_2 gas liberated

3. Compare, Contrast and Classify

Property	Metals	Non-metals	Ionic Compounds
Nature	Generally solids (except Hg – liquid)	Solids, liquids or gases	Crystalline solids
Lustre	Lustrous (shiny)	Dull (non-lustrous)	Generally non-lustrous
Malleability	Malleable – can be beaten into sheets	Not malleable – brittle if solid	Brittle
Ductility	Ductile – can be drawn into wires	Not ductile – cannot be drawn into wires	Not ductile
Conductivity	Good conductors of heat and electricity	Poor conductors (except graphite)	Do not conduct in solid state; conduct when molten or in aqueous solution
Sonority	Sonorous – ringing sound when struck	Non-sonorous – no ringing sound	Non-sonorous
Electron behaviour	Lose electrons to form cations	Gain electrons to form anions	Electrostatic attraction between ions
Oxides	Generally basic; some amphoteric	Generally acidic; some neutral	Formed by reaction of metal oxides with non-metal oxides
Reactivity	Range from highly reactive to less reactive	Generally low	Do not undergo redox reactions like metals
Examples	Fe, Cu, Al, Zn, Na, Ca etc.	O, N, Cl, S, P, I, C (graphite) etc.	NaCl, CaCO_3 , MgO, CuSO_4 etc.

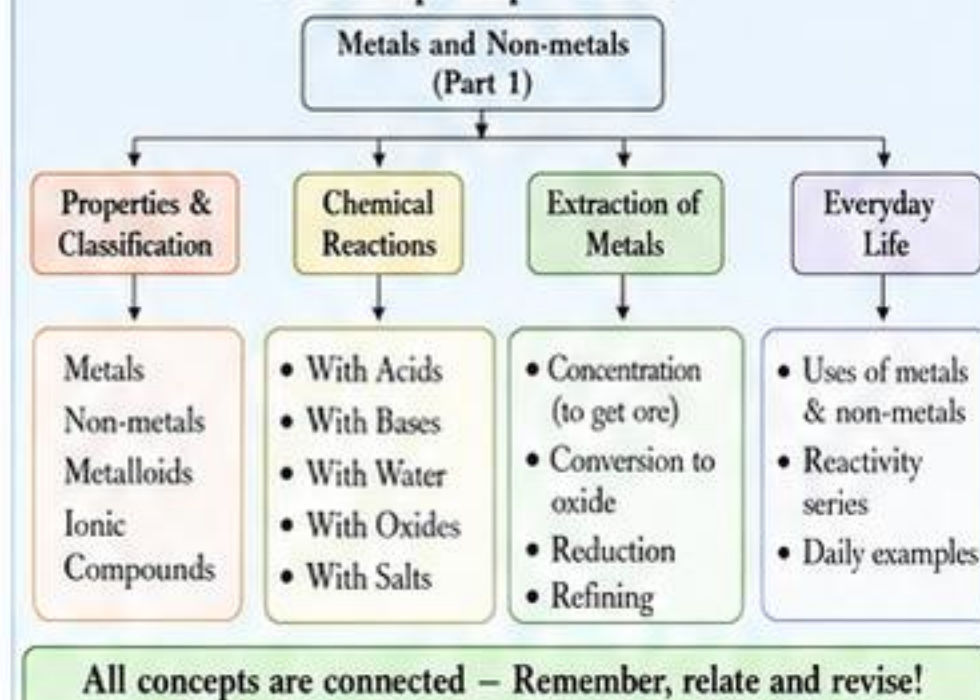
4. Common Misconceptions / NCERT Traps

- ✗ All metals are hard. – False (Sodium, Potassium are very soft).
- ✗ All non-metals are gases. – False (Bromine – liquid, Iodine – solid).
- ✗ Graphite is a non-metal but it conducts electricity.
- ✗ Mercury is a liquid metal at room temperature.
- ✗ Iodine is a non-metal but it is lustrous.
- ✗ All metal oxides are basic. – False (Al_2O_3 , ZnO are amphoteric).
- ✗ All non-metal oxides are acidic. – False (CO , NO are neutral).
- ✗ Sodium and Potassium are stored in kerosene to prevent reaction with air and moisture.
- ✗ Displacement reaction always produces a gas. – False ($\text{Zn} + \text{CuSO}_4$).

5. Short-Reason Answers – High Score Points

- Why ionic compounds conduct electricity only in molten or aqueous state?
Because ions are fixed in solid state; they are free to move only when molten or dissolved in water, allowing current to pass.
- Why are ores converted to oxides before reduction?
Oxides are easier to reduce than sulphides or carbonates and contain more metal per unit mass.
- Why are highly reactive metals extracted by electrolysis?
They are too reactive to be reduced by carbon or even CO ; electrolysis using molten salts gives the pure metal.
- Why is aluminium oxide called amphoteric?
It reacts with both acids (forms aluminium salts + water) and bases (forms sodium aluminate + water).

6. Concept Map – Part 1



7. Part 2 – Coming Next!

Part 2 will continue with:

- ★ Corrosion and its prevention
- ★ Alloys – types, properties and uses
- ★ Important applications of metals and non-metals
- ★ Full Chapter Equation Bank
- ★ Diagram Practice – All important diagrams
- ★ Revision Sheets + Practice Questions + High Score Tips

8. Centum Focus – Finish Strong!

- ★ Revise tables, equations and key definitions daily.
- ★ Learn all reactions with observations and inferences.
- ★ Practice NCERT intext, exercise, exemplar and PYQs.
- ★ Draw diagrams neatly and label correctly.
- ★ Use this sheet for last-minute revision before the exam.

Stay consistent, stay confident, score 100/100! **100**